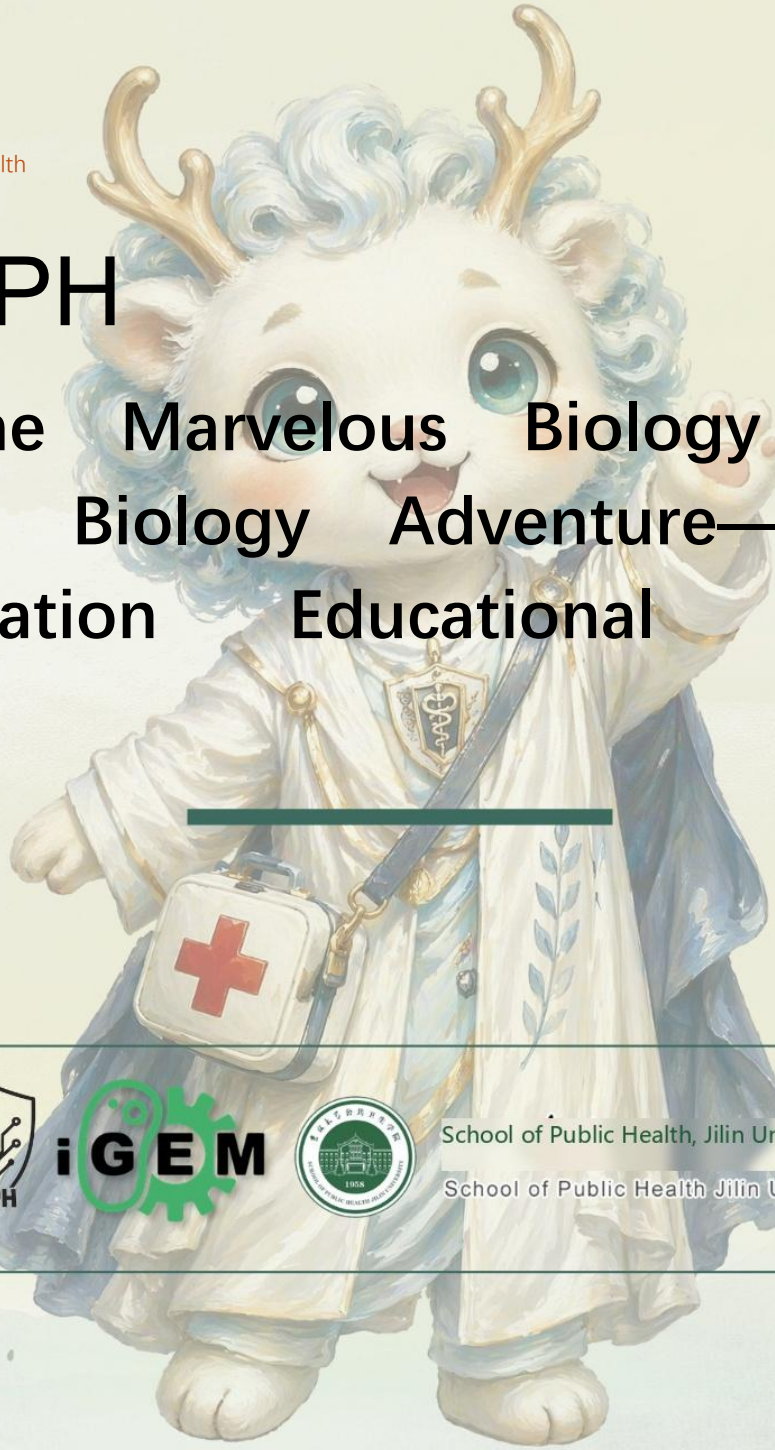
Protecting Life · Public Health

JLU-SPH

2026 The Marvelous Biology LEGO: Synthetic Biology Adventure—Science Popularization Educational Activity Proposal



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I. Basic Information of the Activity

- Activity Title: The Marvelous Bio-Lego: Synthetic Biology Grand Adventure—A Science Popularization Education Program for Youth
- Activity Theme: Designing Life like Building Blocks
- Activity Type: ☑ Science Education ☐ Public Service ☐ Cultural Promotion ☐ Community Service ☐ Others:
- Organizer: JLU-SPH Team

II. Purpose of the Activity

1. Core objective: Centered around the creative concept of "biological LEGO," this program aims to transform specialized concepts such as DNA and gene editing into accessible, easy-to-understand block-building stories. The goal is to help adolescents comprehend the principles, applications, and value of synthetic biology, ignite their interest in exploring life sciences and frontier research, and cultivate scientific thinking and innovation awareness.

2. Specific Objectives:

○ Knowledge/Cognitive Level: Enable children to understand that DNA is the "instruction manual of life." Synthetic biology is likened to "building with LEGO," allowing biological components to be assembled together to create new functions. Students will be introduced to applications of synthetic biology in areas such as bioluminescent plants, advanced materials, pharmaceutical production, and environmental remediation.

○ Behavioral/Skill Level: To cultivate the willingness to think proactively and ask questions, encouraging participants to utilize what they have learned to propose their own creative ideas during the brainstorming session, and to enhance their scientific communication skills.

○ Emotional/Attitudinal Level: To eliminate the sense of distance and apprehension toward frontier sciences, establish a positive attitude that "science is fun and accessible," and stimulate some participants' future interest in life sciences and related disciplines.

3. Expected Outcomes:

○ Coverage: Approximately 40-50 participants.

○ Satisfaction Goal: Over 90%.

○ Achieve full coverage of life sciences and health protection science education for young children.

III. Target Audience

● Core Audience: Preschool children aged 3-5 years.

● Audience Characteristics: Highly curious and imaginative, with an innate interest in life sciences but at an early stage of cognitive development.

IV. Activity Time and Venue

● Event Date: April 27, 2026

● Total Duration: Approximately 40 minutes

● Venue: Kindergarten Children's Handicraft Room

V. Content and Schedule of the Activity

[Overview of Event Flow]

Phase	Time	Session Title	Specific Content	Responsible Personnel	Required Materials/ Media Focus
Phase	10 min	Venue Setup and Participant Assembly	Classroom preparation includes arranging desks and chairs and projection equipment; displaying illustrations and posters analogizing biological LEGO; preparing child-friendly science popularization PPT slides, interactive lecture scripts, and creative imagination record cards. Team members allocate responsibilities and familiarize themselves with lecture scripts and analogical teaching materials.	Overall Coordinator, Logistics	Computer, projector, illustrated posters, banners

Main Session	**10 min**	Section 1: Science Popularization Introduction and DNA of Life Explanation	Commence with the storyline of a “block-building adventure” to quickly capture attention. Introduce the core concepts of life and DNA, using the analogy of DNA as the “instruction manual of life.” This simple analogy elucidates its function in carrying genetic information and determining biological traits.	Lecturer	**PPT**
Main Session	**10 min**	Session 2: Introduction to the Bio-Lego Concept	By introducing the analogy of “building blocks,” the presenter explains how scientists assemble biological parts (gene fragments; English: gene segments) like Lego bricks to create entirely new functions. The comparison between biological modification and synthetic biology is emphasized, highlighting the core advantages of being designable and programmable.	All Teaching Assistants , Lecturer	**PPT**
Main Session	**8 min**	Session 3: Demonstration of Application Scenarios	Illustrate the amazing applications of synthetic biologyin fields such as luminous plants, advanced materials, pharmaceutical production, and environmental remediation. Use concrete, engaging cases like the “biological chef,” "glowing saplings,” and “plastic-eating microbes” to help children intuitively perceive the power of technology. Pose simple questions (e.g., “What would you like to make with Bio-Lego?”).	Lecturer, Teaching Assistants	

Conclusion Phase	**15 min**	Session 4: Interactive Q&A and Creative Brainstorming	Encourage children to raise their hands and ask questions or share their own Bio-Lego ideas (e.g., “trees that can purify air,” “bacteria that can eat garbage”). Team members squat or come closer to listen attentively and give positive feedback.	All Team Members	**PPT**

[Detailed Explanation of Each Session]

Section 1: Science Popularization Introduction and DNA of Life Explanation

- **Core content:** The session starts with the narrative framework of a "block-building adventure" (e.g., "Children, today we are going to explore a magical kingdom of life with building blocks!"). Using child-friendly language, the lesson explains what life is and further analogizes DNA as the "manual of life"—like a thick book that describes what we look like and what abilities we possess. Emphasis is placed on the fact that all living beings possess this manual.
- **Key phrases:** "Children, did you know? Each of us has a magical 'manual of life' inside our bodies, which determines what we look like, the color of our eyes, and whether we can run fast! This manual is called DNA!" "Plants and small animals also have their own DNA manuals!"

Session 2: Introduction to the Bio-Lego Concept

- **Core Content:** By displaying physical Lego bricks or pictures, guide participants to recall their experiences with building blocks—how different pieces can form houses, cars, and robots. Then use analogy to explain: scientists use different gene fragments just like "Lego pieces," combining them to create organisms with new functions! This is called "Synthetic Biology." Emphasize that it is "designable" and "creatable."
- **Key Talking Points:** "Has everyone played with Lego before? Different building blocks can be assembled into all sorts of things. Scientists do something similar by assembling different gene fragments to create organisms with new functions. Isn't that cool?" "This is called Synthetic Biology—designing life like building with Lego!"

VI. Staffing and Division of Responsibilities

- **General Supervisor (1 person):** Oversees coordination, liaises with the kindergarten, manages process and timing, and addresses emergencies. Must have strong communication skills and comprehensive situational awareness of the activity.
- **Main Lecturer (1 person):** Responsible for explaining and demonstrating Sessions 1, 2, and 3. Should be expressive, engaging, and adept at using analogies and storytelling to clarify abstract concepts for children.
- **Onsite Teaching Assistants (3 persons):** Responsible for registration, distributing materials, and maintaining order. Should be patient, attentive, and experienced in caring for young children.
- **Photography and Logistics: 2 personne:** Responsible for inventory management, site cleaning, and emergency support.

VII. List of Materials

- Equipment:
  - Notebook computer, projector

- ○ Camera/high-definition mobile phone
- ● Teaching materials:
- ○ Science popularization PPT slides
- ○ Activity feedback questionnaire
- ● Logistics:
- ○ Standard first aid kit
- ○ Cleaning wet wipes, garbage bags

VIII. Risk Contingency Plan

- Attention distraction or disorder: Communicate with school teachers prior to the activity and ensure teacher presence throughout to assist in maintaining order; implement a compact presentation pace, with each session not exceeding 12 minutes; utilize interesting images and short videos (if available) during scenario demonstrations to enhance novelty; teaching assistants monitor students seated in the back or displaying inattentiveness and provide timely reminders as necessary.
- Low participation during Q&A or brainstorming sessions: Team members prepare 2–3 exemplary answers in advance (e.g., "I want to design a tree that produces bread") to proactively stimulate discussion during lulls; the general coordinator or instructor may begin by sharing their own creative idea in the "if I were a designer" theme to lower the psychological threshold for children to express themselves.

IX. Effectiveness Evaluation

- Process assessment: Through on-site Q&A (“What is DNA?” “What is synthetic biology like?”) and creative idea-sharing sessions, the quantity and content of student responses will be recorded to evaluate understanding of core concepts and the degree of interest in science that has been stimulated.
- Satisfaction assessment: Distribute a brief questionnaire to the accompanying teachers to collect feedback on the content, format, and organization of the activity; collect verbal feedback from children regarding their participation experience.